PYTHON PROGRAMMING MID-TERM PROJECT

SSCMS — Survivor Support and Case Management System

Track	Track 2 — Desktop App (GUI)
External library	customtkinter 5.2.2
Persistence	Local JSON storage with CSV export capability

Repository: https://github.com/ZobayerJony/Python_Project_Mid-Term-SSCMS

1. Executive Summary

SSCMS is a Python desktop application for managing survivor support cases through a graphical user interface built with customtkinter. The system supports case creation, review, updating, deletion, search, multi-criteria filtering, sorting, reporting, timeline-based follow-up tracking, and CSV export. The repository is organized into clear layers for models, storage, business logic, services, and UI code, which makes the project easy to explain and suitable for a mid-term submission.

Main entry point	main.py
Core package	sscms/
Primary post-login views	Cases, Add/Edit Case, Case Details, Reports, Export
Data file used by the app	data/data.json
Additional export output	data/exports/sscms_cases_<timestamp>.csv

2. Mandatory Setup and Run Instructions

Step	Command	Purpose
1	python -m venv .venv	Create the project virtual environment.
2	.venv\Scripts\activate	Activate the environment on Windows.
3	source .venv/bin/activate	Activate the environment on macOS/Linux.
4	pip install -r requirements.txt	Install external dependencies.
5	python main.py	Launch the SSCMS application.
6	pip freeze > requirements.txt	Freeze dependencies for submission.

3. Requirement Compliance Matrix

Requirement	What the brief expects	Project evidence	Verdict
Desktop App track	A record-management GUI app using customtkinter, PySimpleGUI, or tkinter.	requirements.txt; sscms/ui/; multiple CTk-based view files.	Satisfied

External library	At least one pip-installed external package.	customtkinter imported across the UI layer; dependencies listed in requirements.txt.	Satisfied
At least 2 views/screens	More than one interface screen or page.	Login, Cases, Add/Edit Case, Case Details, Reports, and Export views.	Satisfied
CRUD operations	Add, view, edit, and delete records.	CaseManager.add_case(), get_by_id(), update_case(), delete_case(); CasesView and CaseFormView buttons.	Satisfied
Search and sort/filter	Search plus sorting or filtering tools.	CaseManager.search(), filter_cases(), sort_cases(); CasesView search bar and dropdown filters.	Satisfied
Summary report	Totals, averages, top items, or similar reporting.	CaseManager.summary_report(); ReportsView report screen.	Satisfied
Persistence	Data saved and loaded after restart.	JsonStore load/save logic with data/data.json storage.	Satisfied
OOP design	At least 4 classes with clear roles.	14 classes across models, manager, storage, services, and UI.	Satisfied
Validation and robustness	Graceful handling of invalid input and edge cases.	Field validation helpers, message boxes, empty-result handling, JSON backup fallback.	Satisfied
Environment + packaging	venv workflow, requirements file, and run steps shown in report.	requirements.txt present; commands documented in this report and README_REPORT.md.	Satisfied

4. Project Structure

```
SSCMS/
|-- main.py
|-- requirements.txt
|-- README_REPORT.md
|-- data/
|   |-- data.json
|   \-- exports/
\-- sscms/
    |-- __init__.py
    |-- config.py
    |-- models.py
    |-- storage.py
    |-- manager.py
    |-- services.py
    |-- utils.py
    \-- ui/
        |-- app.py
        |-- login_view.py
        |-- theme.py
        |-- widgets.py
        \-- views/
            |-- cases_view.py
            |-- case_form_view.py
            |-- case_detail_view.py
            |-- reports_view.py
            \-- export_view.py
```

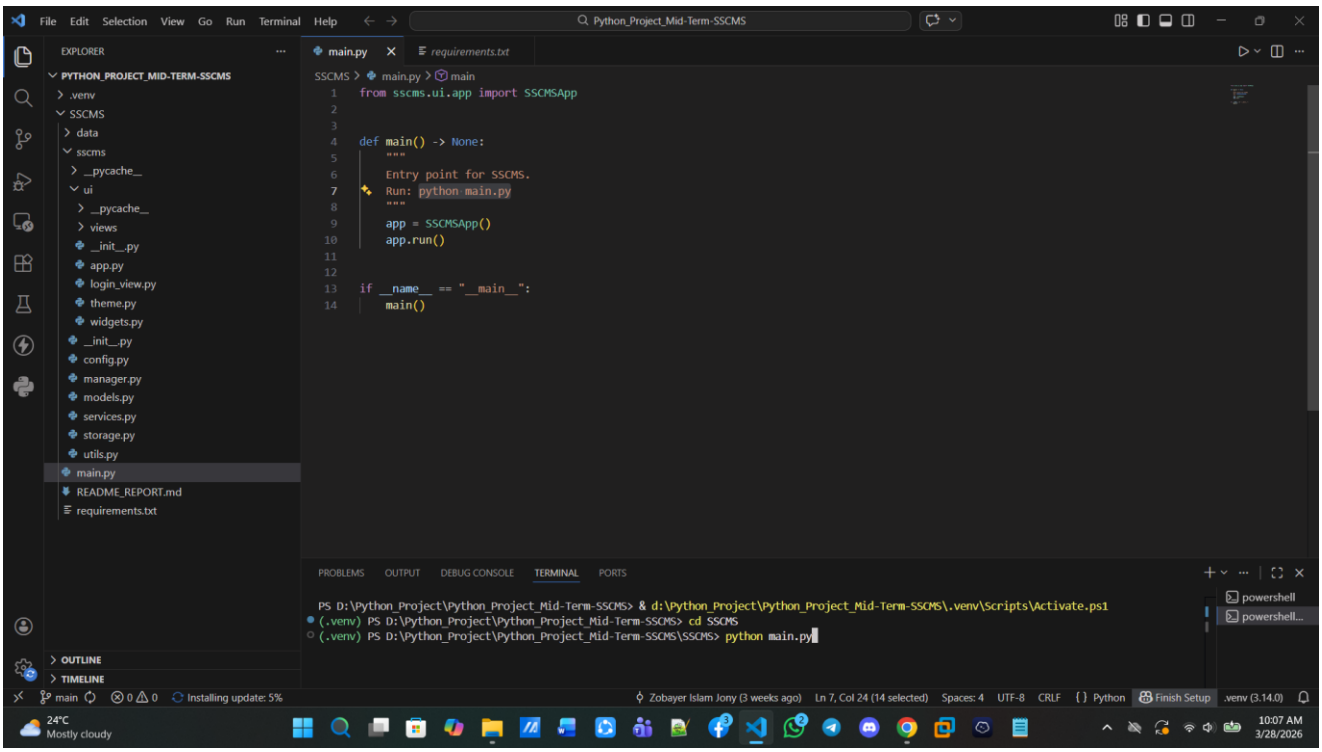
Core/data file	Role in the project
main.py	Entry point that creates SSCMSApp and starts the program.
requirements.txt	Lists the packages needed to run the GUI application.
data/data.json	Stores persisted case data and the next available case ID.
sscms/config.py	Defines paths, dropdown values, window settings, and login constants.
sscms/models.py	Defines SurvivorCase and CaseActivity entity models.
sscms/storage.py	Loads and saves JSON data, including default-file creation and backup handling.
sscms/manager.py	Contains validation, CRUD, search, filter/sort, and summary-report logic.
sscms/services.py	Exports case records to timestamped CSV files.
sscms/utils.py	Holds reusable validation helpers such as phone and date checks.

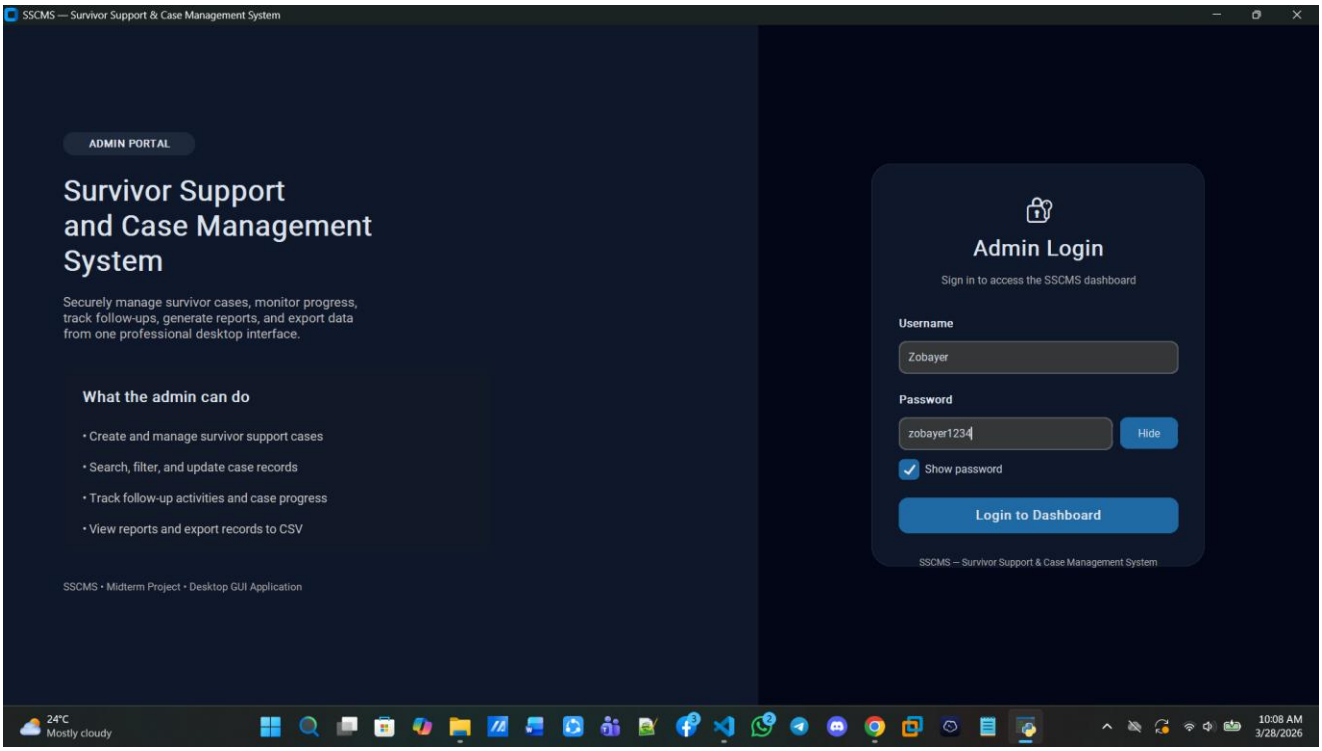
UI file	Role in the project
sscms/ui/app.py	Builds the window shell, sidebar, content area, status bar, and view navigation.
sscms/ui/login_view.py	Implements the admin login screen shown before the main dashboard.
sscms/ui/theme.py	Applies the default appearance mode, color theme, and shared fonts.
sscms/ui/widgets.py	Provides reusable SectionHeader and StatusBar widgets.
sscms/ui/views/cases_view.py	Displays the main cases table with search, filter, sort, refresh, edit, and delete actions.
sscms/ui/views/case_form_view.py	Handles both add-case and edit-case workflows.
sscms/ui/views/case_detail_view.py	Shows full case details, status changes, and timeline follow-up notes.

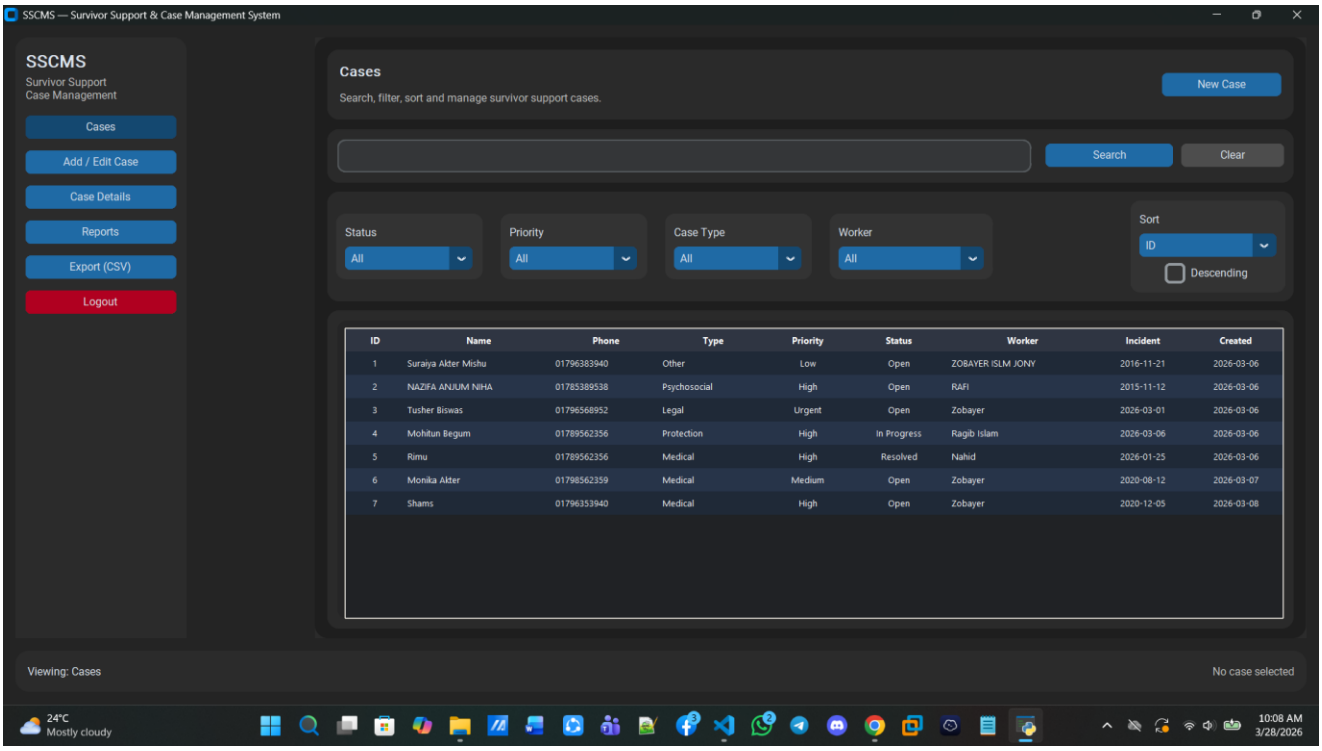
sscms/ui/views/reports_view.py	Formats the summary report for the user.
sscms/ui/views/export_view.py	Runs CSV export from the GUI.

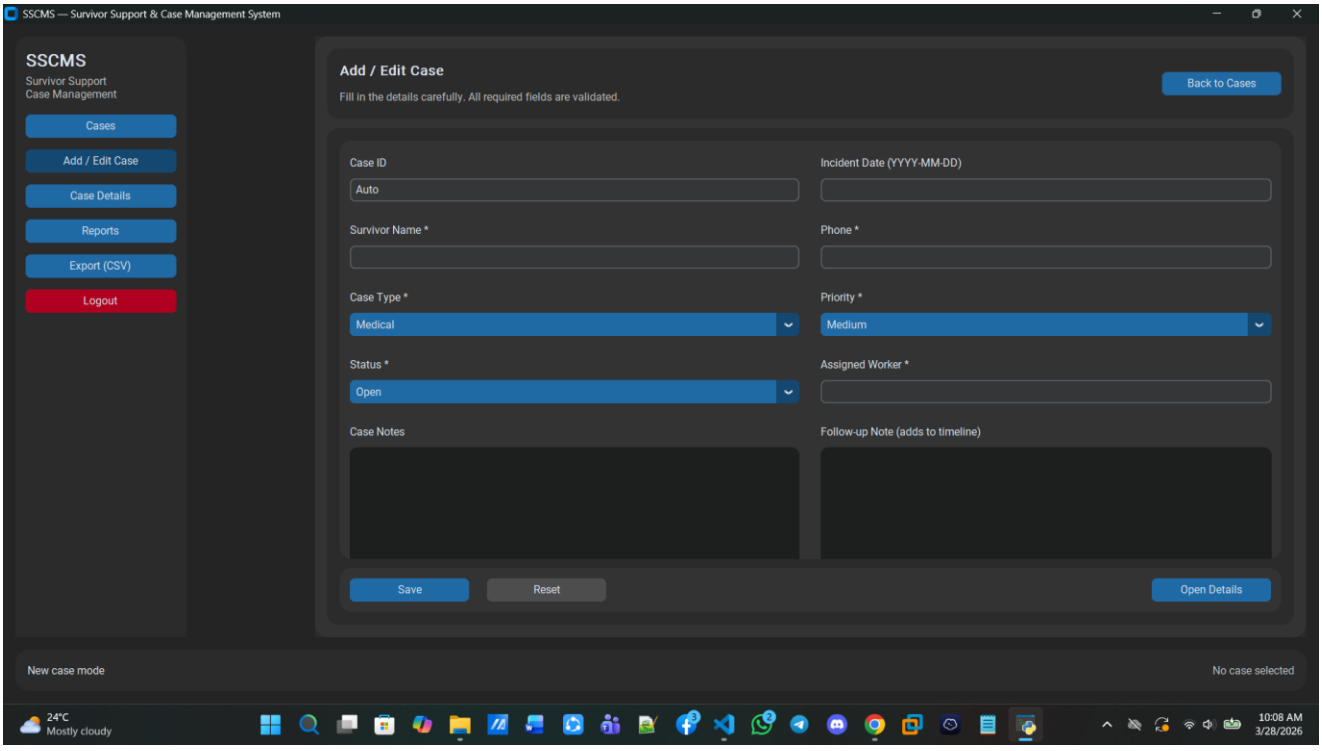
1. The user runs python main.py, which imports SSCMSApp and starts the program.
2. SSCMSApp applies the theme, creates the main window, and initializes CaseManager(JsonStore(DATA_FILE)) plus ExportService.
3. The app first shows LoginView, which checks the admin username and password before giving access to the dashboard.
4. After login, SSCMSApp builds the sidebar, content area, and status bar, then initializes all views.
5. CasesView reads the current records through CaseManager and displays them in a styled table.
6. CaseFormView sends add or edit actions to add_case() and update_case(), which validate input and persist changes.
7. CaseDetailView reads the selected case, shows the timeline, allows quick status changes, and appends follow-up activities.
8. ReportsView calls summary_report() to display totals, breakdowns, and averages.
9. ExportView calls ExportService.export_cases_csv() and writes a timestamped CSV file into data/exports/.
10. Every successful data change is written back to data/data.json through JsonStore, so the data survives restart.

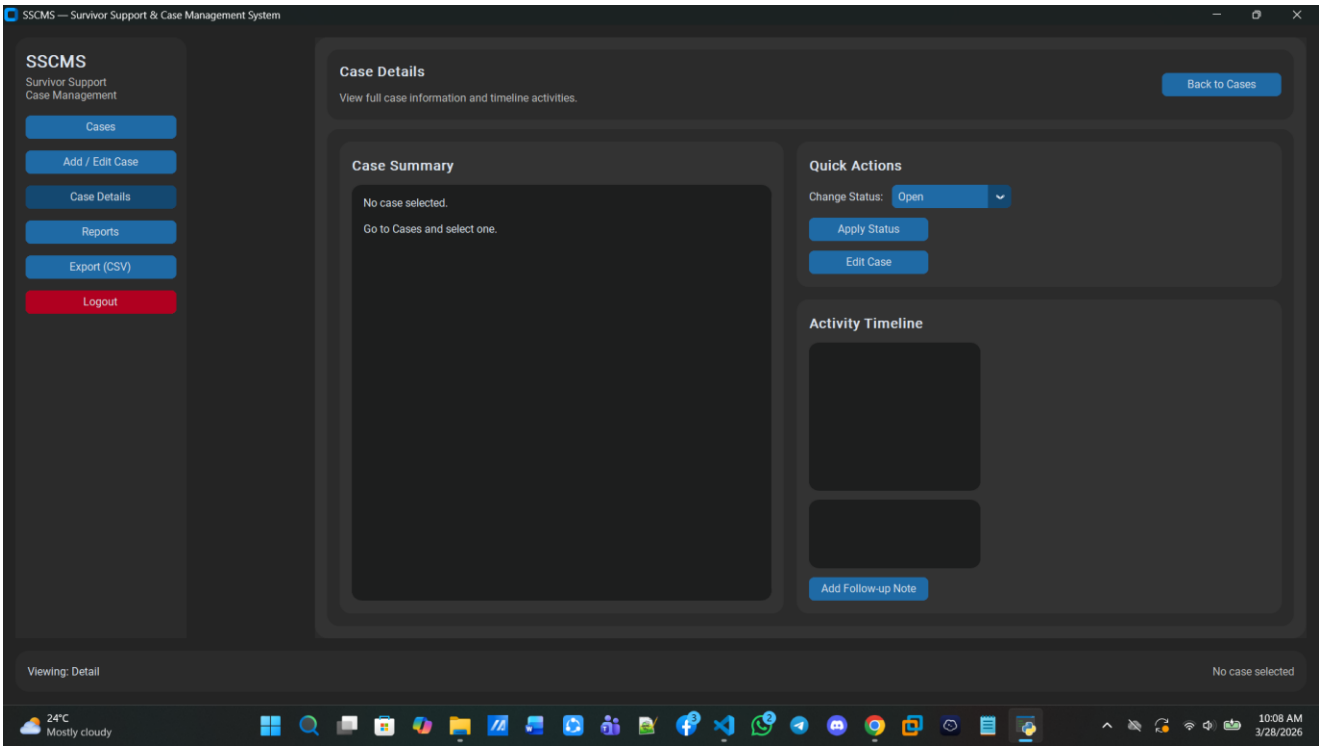
5. Run Instructions And Screenshots:

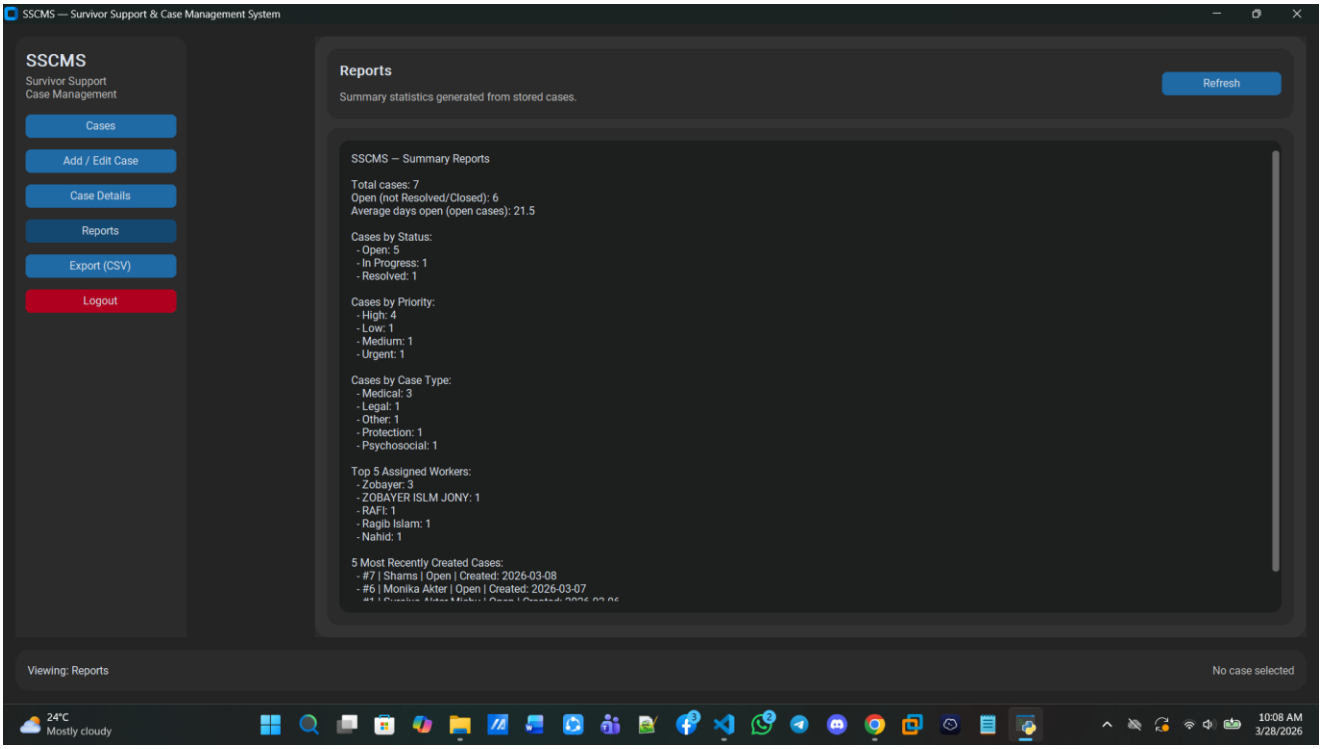


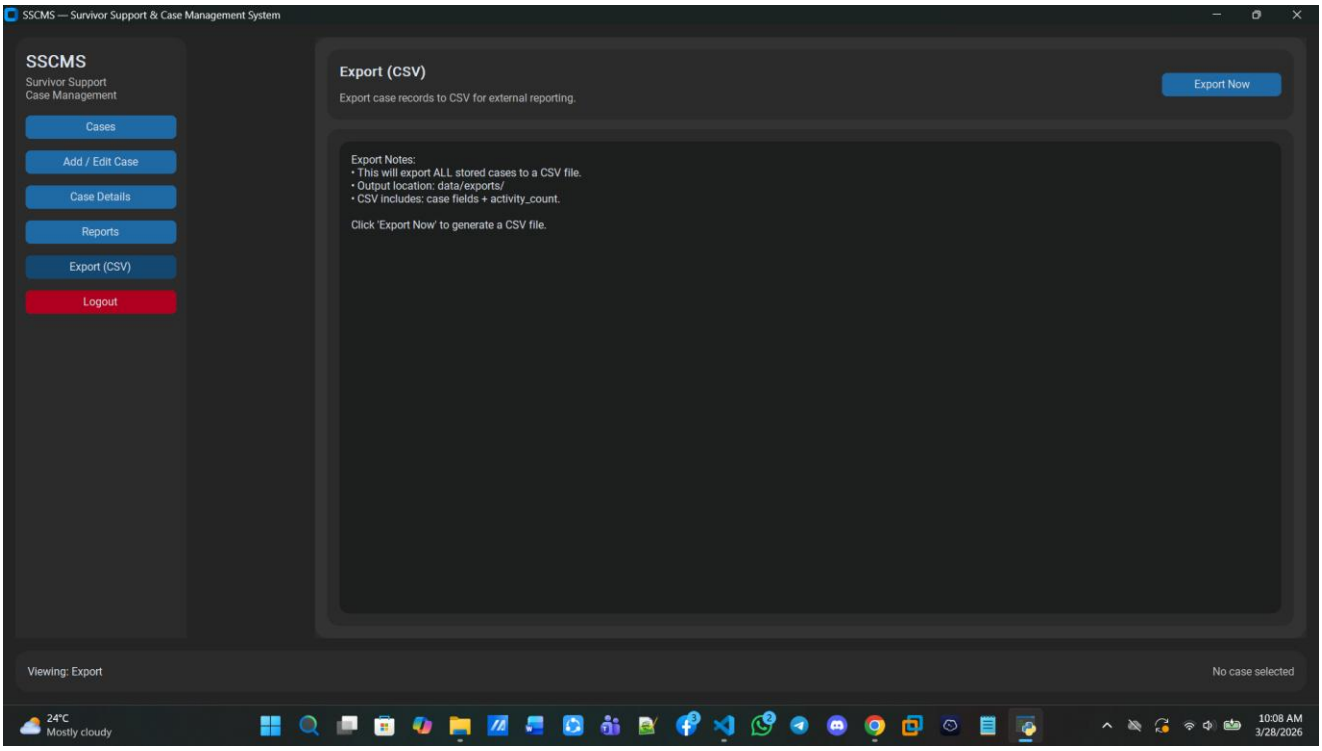


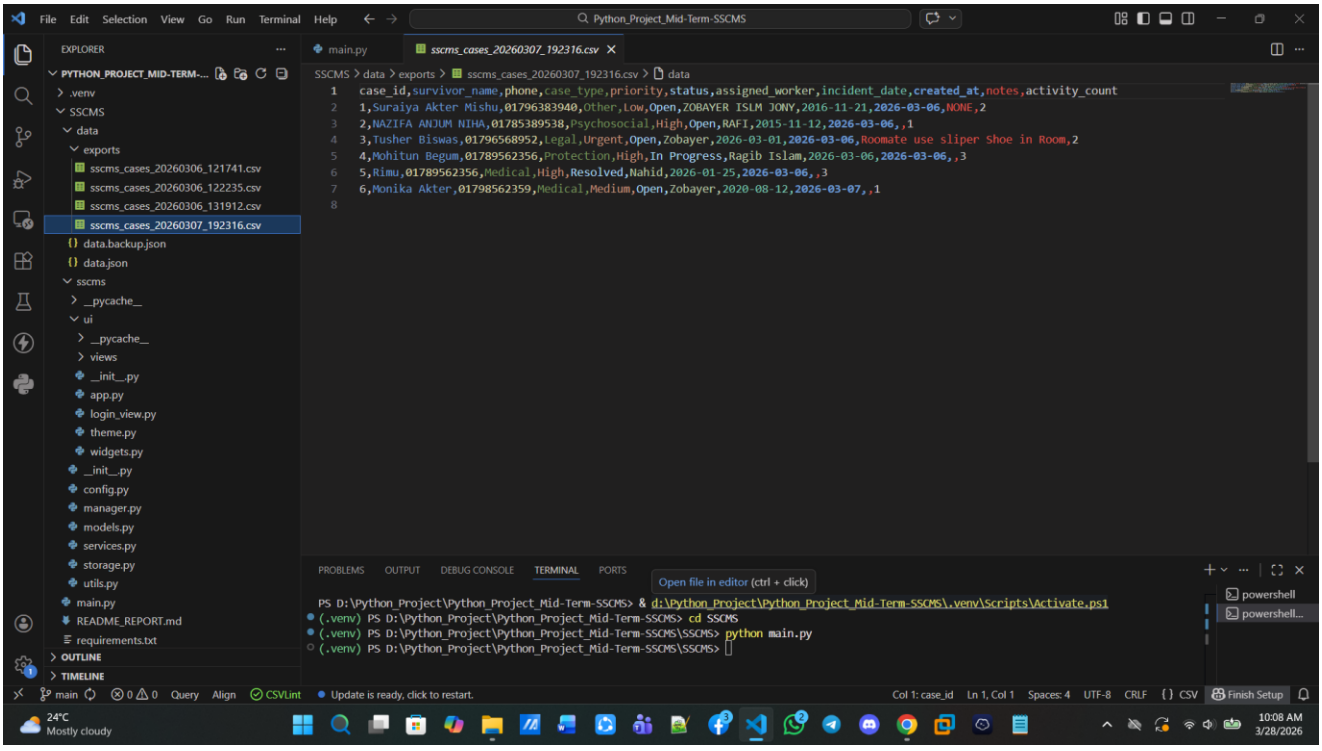

SSCMS — Survivor Support & Case Management System
SSCMS
Survivor Support
Case Management
Cases
Add / Edit Case
Case Details
Reports
Export (CSV)
Logout
Cases
Search, filter, sort and manage survivor support cases.
New Case
Search
Clear
Status
All
Priority
All
Case Type
All
Worker
All
Sort
ID
Descending
ID Name Phone Type Priority Status Worker Incident Created
1 Suraiya Akter Mishu 01796383940 Other Low Open ZOBAYER ISLM JONY 2016-11-21 2026-03-06
2 NAZIFA ANJUM NIHA 01785389538 Psychosocial High Open RAFI 2015-11-12 2026-03-06
3 Tusher Biswas 01796568952 Legal Urgent Open Zobayer 2026-03-01 2026-03-06
4 Mohitun Begum 01789562356 Protection High In Progress Ragib Islam 2026-03-06 2026-03-06
5 Rimu 01789562356 Medical High Resolved Nahid 2026-01-25 2026-03-06
6 Monika Akter 01798562359 Medical Medium Open Zobayer 2020-08-12 2026-03-07
7 Shams 01796353940 Medical High Open Zobayer 2020-12-05 2026-03-08
Viewing: Cases
No case selected
24°C
Mostly cloudy
10:08 AM
3/28/2026


SSCMS — Survivor Support & Case Management System
SSCMS
Survivor Support
Case Management
Cases
Add / Edit Case
Case Details
Reports
Export (CSV)
Logout
Add / Edit Case
Fill in the details carefully. All required fields are validated.
Back to Cases
Case ID
Auto
Incident Date (YYYY-MM-DD)
Survivor Name *
Phone *
Case Type *
Medical
Priority *
Medium
Status *
Open
Assigned Worker *
Case Notes
Follow-up Note (adds to timeline)
Save
Reset
Open Details
New case mode
No case selected
24°C
Mostly cloudy
10:08 AM
3/28/2026


SSCMS — Survivor Support & Case Management System
SSCMS
Survivor Support
Case Management
Cases
Add / Edit Case
Case Details
Reports
Export (CSV)
Logout
Case Details
View full case information and timeline activities.
Back to Cases
Case Summary
No case selected.
Go to Cases and select one.
Quick Actions
Change Status:
Open
Apply Status
Edit Case
Activity Timeline
Add Follow-up Note
Viewing: Detail
No case selected
24°C
Mostly cloudy
10:08 AM
3/28/2026


SSCMS — Survivor Support & Case Management System
SSCMS
Survivor Support
Case Management
Cases
Add / Edit Case
Case Details
Reports
Export (CSV)
Logout
Reports
Summary statistics generated from stored cases.
Refresh
SSCMS — Summary Reports
Total cases: 7
Open (not Resolved/Closed): 6
Average days open (open cases): 21.5
Cases by Status:
- Open: 5
- In Progress: 1
- Resolved: 1
Cases by Priority:
- High: 4
- Low: 1
- Medium: 1
- Urgent: 1
Cases by Case Type:
- Medical: 3
- Legal: 1
- Other: 1
- Protection: 1
- Psychosocial: 1
Top 5 Assigned Workers:
- Zobayer: 3
- ZOBAYER ISLM JONY: 1
- RAFI: 1
- Ragib Islam: 1
- Nahid: 1
5 Most Recently Created Cases:
- #7 | Shams | Open | Created: 2026-03-08
- #6 | Monika Akter | Open | Created: 2026-03-07
Viewing: Reports
No case selected
24°C
Mostly cloudy
10:08 AM
3/28/2026


SSCMS — Survivor Support & Case Management System
SSCMS
Survivor Support
Case Management
Cases
Add / Edit Case
Case Details
Reports
Export (CSV)
Logout
Export (CSV)
Export case records to CSV for external reporting.
Export Now
Export Notes:
• This will export ALL stored cases to a CSV file.
• Output location: data/exports/
• CSV includes: case fields + activity_count.
Click 'Export Now' to generate a CSV file.
Viewing: Export
No case selected
24°C
Mostly cloudy
10:08 AM
3/28/2026


File Edit Selection View Go Run Terminal Help
Python_Project_Mid-Term-SSCMS
EXPLORER
PYTHON_PROJECT_MID-TERM-...
.venv
SSCMS
data
exports
sscms_cases_20260306_121741.csv
sscms_cases_20260306_122235.csv
sscms_cases_20260306_131912.csv
sscms_cases_20260307_192316.csv
data.backup.json
data.json
sscms
__pycache__
ui
__pycache__
views
__init__.py
app.py
login_view.py
theme.py
widgets.py
__init__.py
config.py
manager.py
models.py
services.py
storage.py
utils.py
main.py
README_REPORT.md
requirements.txt
OUTLINE
TIMELINE
main.py
sscms_cases_20260307_192316.csv
SSCMS > data > exports > sscms_cases_20260307_192316.csv > data
case_id,survivor_name,phone,case_type,priority,status,assigned_worker,incident_date,created_at,notes,activity_count
1,Suraiya Akter Mishu,01796383940,Other,Low,Open,ZOBAYER ISLM JONY,2016-11-21,2026-03-06,NONE,2
2,NAZIFA ANJUM NIHA,01785389538,Psychosocial,High,Open,RAFI,2015-11-12,2026-03-06,,1
3,Tusher Biswas,01796568952,Legal,Urgent,Open,Zobayer,2026-03-01,2026-03-06,Roomate use sliper Shoe in Room,2
4,Mohitun Begum,01789562356,Protection,High,In Progress,Ragib Islam,2026-03-06,2026-03-06,,3
5,Rimu,01789562356,Medical,High,Resolved,Nahid,2026-01-25,2026-03-06,,3
6,Monika Akter,01798562359,Medical,Medium,Open,Zobayer,2020-08-12,2026-03-07,,1
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Open file in editor (ctrl + click)
PS D:\Python_Project\Python_Project_Mid-Term-SSCMS> & d:\Python_Project\Python_Project_Mid-Term-SSCMS\.venv\Scripts\Activate.ps1
(.venv) PS D:\Python_Project\Python_Project_Mid-Term-SSCMS> cd SSCMS
(.venv) PS D:\Python_Project\Python_Project_Mid-Term-SSCMS\SSCMS> python main.py
(.venv) PS D:\Python_Project\Python_Project_Mid-Term-SSCMS\SSCMS>
powershell
powershell...
main
Query
Align
CSVLint
Update is ready, click to restart.
Col 1: case_id
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
{} CSV
Finish Setup
24°C
Mostly cloudy
10:08 AM
3/28/2026